Number

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

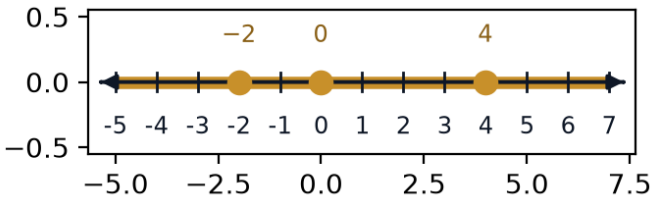
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Estimation

In a class of 30 students, 15 students play football and 20 students play basketball. Every student plays at least one of the two sports. Using a Venn diagram, find the number of students who play both football and basketball.

2 [Foundational] • Target: Rounding

A number line is shown below. What is the value of the number indicated by the arrow?



3 [Foundational] • Target: Four operations with whole numbers

The histogram shows the distribution of scores for a group of students in a mathematics test. The total number of students who scored between 0 and 40 is 32. The frequency density for the class interval $20 \leq x < 40$ is 0.8. Calculate the number of students who scored between 20 and 40.

4 [Foundational] • Target: Order of operations with whole numbers

The diagram shows a number line. The distance between the points marked A and B is 15 units. The coordinate of A is -7. Find the possible coordinates of B.

5 [Foundational] • Target: Using technology for operations with whole numbers

A bag contains n marbles, of which 4 are green. Two marbles are taken at random without replacement. If the probability that both are green is $\frac{2}{15}$, find the total number of marbles n.

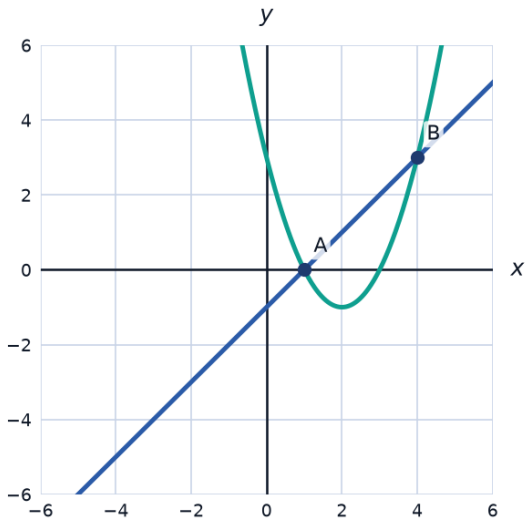
6 [Foundational] • Target: Check reasonableness of calculations

Set A has 12 elements and set B has 8 elements. If $A \cap Beq\varnothing$, what is the maximum possible number of elements in $A \cup B$?

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7 [Foundational] • Target: Estimation

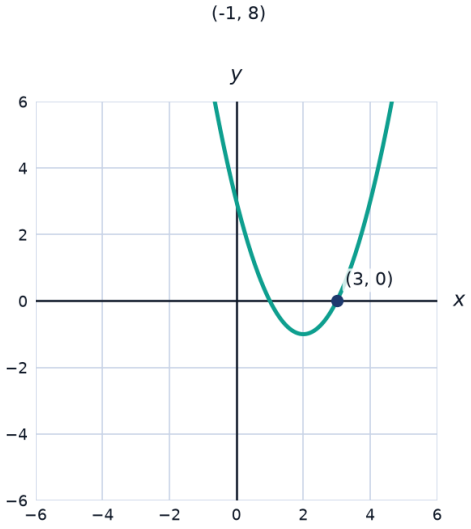
The graph of the quadratic function $y = x^2 - 4x + 3$ is shown. The graph of the linear function $y = x - 1$ is also shown. The graphs intersect at points A and B. Find the coordinates of point A.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Rounding

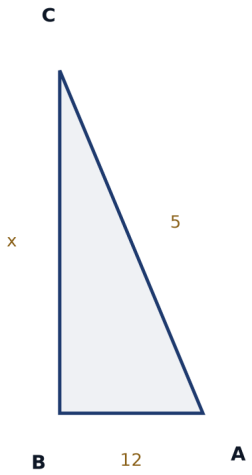
The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.



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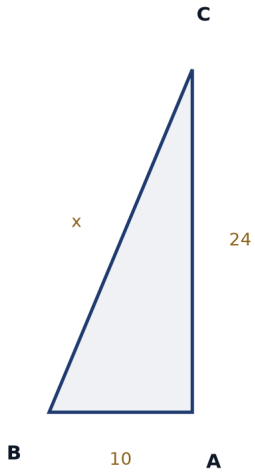
9 **[Intermediate]** • **Target: Four operations with whole numbers**

In the right-angled triangle ABC, $\angle ABC = 90^\circ$. If $AB = 5$ cm and $BC = 12$ cm, find the length of the hypotenuse AC.



10 **[Intermediate]** • **Target: Order of operations with whole numbers**

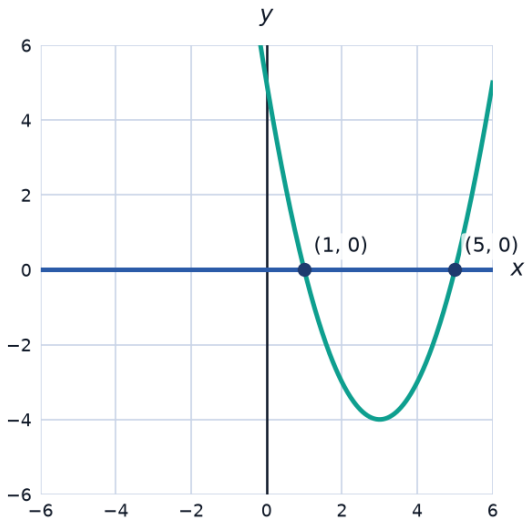
The diagram shows a right-angled triangle ABC, with the right angle at B. The length of side AB is 10 cm and the length of side BC is 24 cm. A semicircle is drawn with AC as its diameter. Calculate the area of the semicircle in terms of π.



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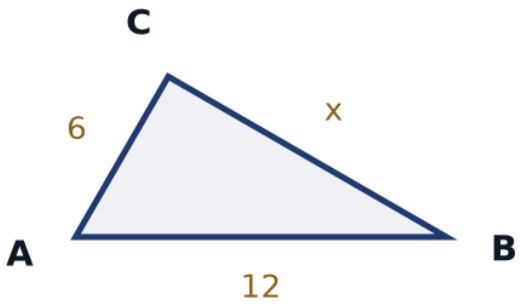
11 [Intermediate] • Target: Using technology for operations with whole numbers

The diagram shows the graph of the quadratic function $y = x^2 + bx + c$. The graph intersects the x-axis at $(1, 0)$ and $(5, 0)$. Find the values of b and c.



12 [Intermediate] • Target: Check reasonableness of calculations

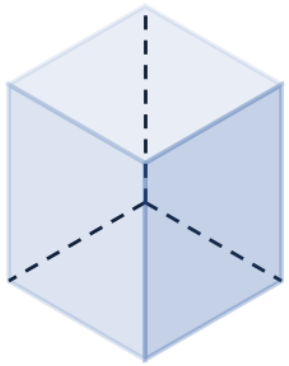
In a right-angled triangle, one of the acute angles is 30° and the side opposite to this angle is 6 cm. Calculate the length of the hypotenuse.



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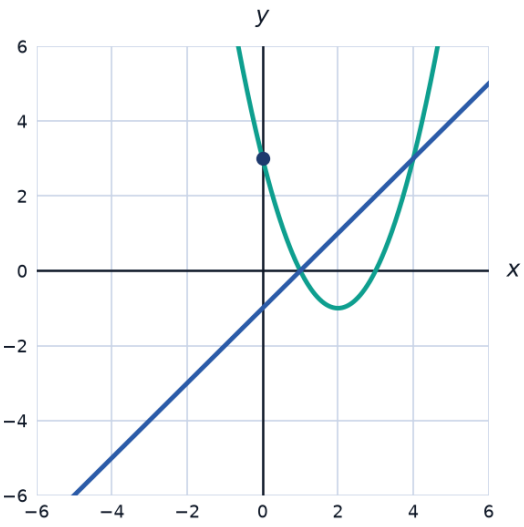
13 [Intermediate] • Target: Estimation

A rectangular prism has a length of 5 cm, a width of 3 cm, and a height of 4 cm. Calculate the volume of the prism.



14 [Intermediate] • Target: Rounding

The graph of the function $y = f(x)$ is shown below. Find the value of $f(0)$.

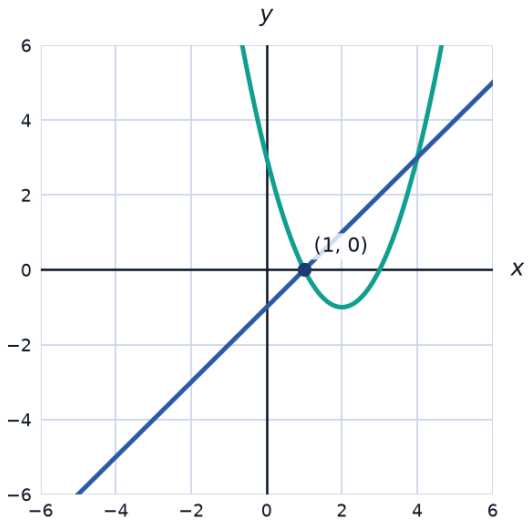


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Four operations with whole numbers • Cross-Lesson: 6.01 Trigonometry

The diagram shows the graph of the quadratic function $y = x^2 - 4x + k$. The graph intersects the x-axis at point P(1, 0) and the y-axis at point Q. Determine the value of k.



16 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 7.01 Transformations & Vectors

In the diagram below, lines AB and DE are parallel. Points A, C, and D are collinear, and points B, C, and E are collinear. If $AC = 6$ cm, $CD = 9$ cm, and $DE = 12$ cm, find the length of AB.

17 [Challenging] • Target: Using technology for operations with whole numbers • Cross-Lesson: 8.01 Probability & Statistics

A point $A(3, 5)$ is mapped to $A'(1, 3)$ by a single transformation. The same transformation is then applied to a second point $B(x, y)$, mapping it to $B'(x - 2, y - 2)$. The vector representing the displacement from A to A' is $\vec{v}$. A point $C(2, 1)$ is reflected in the line $y = x$ to give point C'. Calculate the magnitude of the vector $\vec{AC'}$.

18 [Challenging] • Target: Check reasonableness of calculations • Cross-Lesson: 9.01 Functions

A survey of 100 students about their preferred social media platforms yielded the following information: - 55 students use Platform A. - 62 students use Platform B. - 18 students use neither Platform A nor Platform B. - x students use both Platform A and Platform B. Find the value of x.

19 [Challenging] • Target: Estimation • Cross-Lesson: 10.01 Sets & Venn Diagrams

A triangle ABC has vertices A(2, 3), B(5, 1), and C(3, 5). This triangle is subjected to a sequence of transformations. First, it is reflected in the line $y = x$ to produce triangle A'B'C'. Then, triangle A'B'C' is translated by the vector $(-3; 1)$ to produce triangle A"B"C". ; Determine the coordinates of the vertices of triangle A"B"C".

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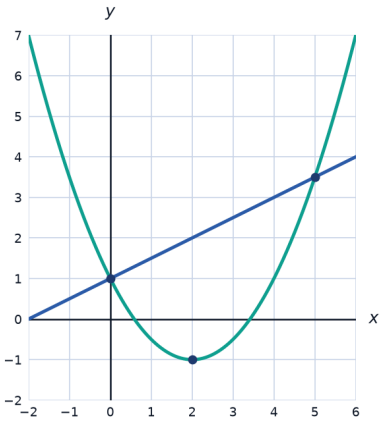
20 [Challenging] • Target: Rounding • Cross-Lesson: 2.01 Algebra & Graphs

The diagram shows a point A and its image A' after a single transformation. The coordinates of A are (2, 1) and the coordinates of A' are (5, 3). The transformation can be represented by the vector $(p; q)$. Find the value of p.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Estimation

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(2)$.



a. 7

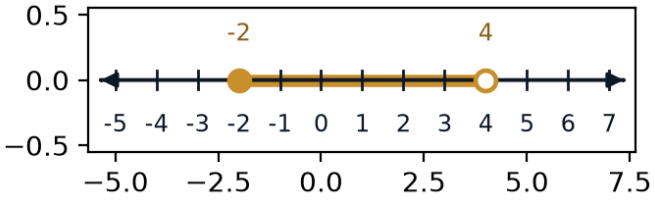
b. 9

c. 4

d. 11

22 [Foundational] • Target: Estimation

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

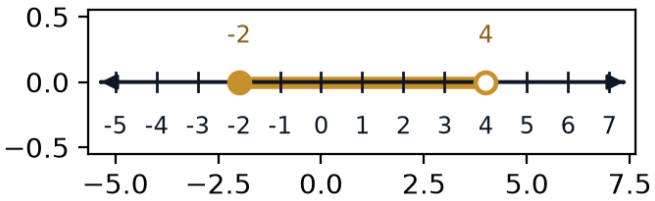
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Estimation

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(5)$.



a. 13

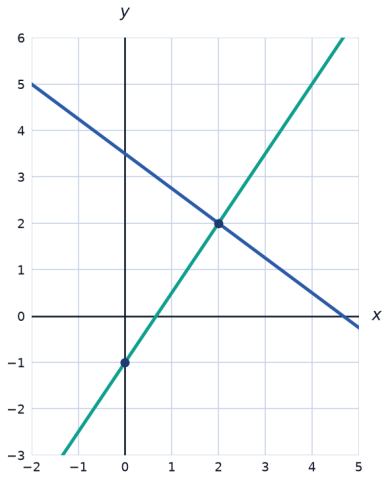
b. 15

c. 10

d. 17

24 [Foundational] • Target: Rounding

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(2)$.



a. 10

b. 13

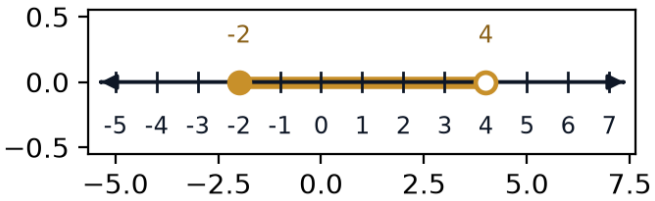
c. 6

d. 14

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25 [Foundational] • Target: Rounding

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

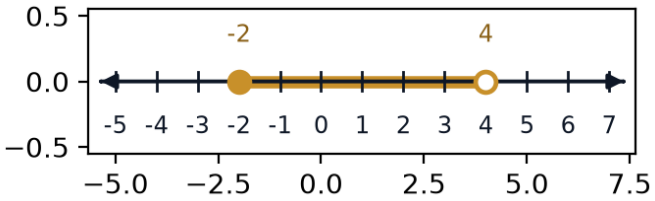
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Rounding

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(5)$.



a. 19

b. 22

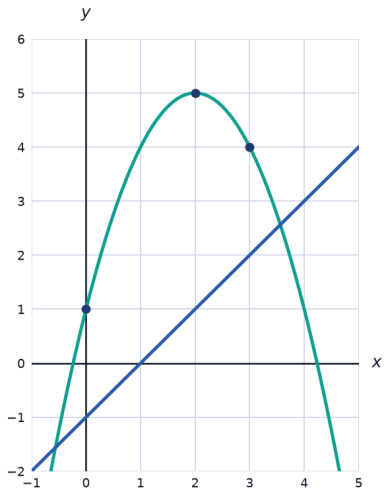
c. 15

d. 23

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27 [Foundational] • Target: Four operations with whole numbers

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(2)$.



a. 13

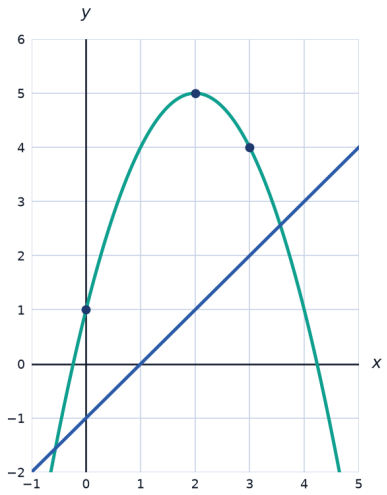
b. 17

c. 8

d. 17

28 [Foundational] • Target: Four operations with whole numbers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Estimation

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Estimation

A data set recorded for estimation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

31 [Intermediate] • Target: Estimation

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(4)$.

a. 11

b. 13

c. 8

d. 15

32 [Intermediate] • Target: Estimation

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

33 [Intermediate] • Target: Rounding

A data set recorded for rounding consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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34 [Intermediate] • Target: Rounding

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

35 [Intermediate] • Target: Rounding

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

36 [Intermediate] • Target: Four operations with whole numbers

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

37 [Intermediate] • Target: Four operations with whole numbers

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Four operations with whole numbers • Cross-Lesson: 9.01 Functions

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(4)$.

a. 21

b. 25

c. 16

d. 25

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39 [Challenging] • Target: Four operations with whole numbers • Cross-Lesson: 10.01 Sets & Venn Diagrams

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 2.01 Algebra & Graphs

A data set recorded for order of operations with whole numbers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

41 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 5x + 6$ in order of operations with whole numbers, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

42 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 4.01 Mensuration

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

43 [Challenging] • Target: Using technology for operations with whole numbers • Cross-Lesson: 5.01 Geometry

A data set recorded for using technology for operations with whole numbers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

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44 [Challenging] • Target: Using technology for operations with whole numbers • Cross-Lesson: 6.01 Trigonometry

For the model $f(x) = 6x + 7$ in using technology for operations with whole numbers, evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

45 [Challenging] • Target: Check reasonableness of calculations • Cross-Lesson: 7.01 Transformations & Vectors

For the model $f(x) = 2x + 8$ in check reasonableness of calculations, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

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